

Benjamin Blair

Junior-level Computer Science Undergraduate, Class of 2027 | Founder of JesusClubCU.com | Dana Point, CA 92677
Email: ben0r0blair@gmail.com | GitHub: github.com/Ben-Blair | GPA: 3.60

Summary

Junior-level Computer Science undergraduate with hands-on experience in machine learning, data analysis, and software development. Skilled in building and deploying models using Python, OpenCV, and Scikit-Learn, with practical experience in real-time data processing and modular system design. I really like building cool projects and would love to get my hands dirty with real-life work experience!

Education

University of Colorado Boulder — Bachelor of Science in Computer Science
August 2024 – May 2027

Top Projects

Real-Time Video Monitoring & Detection System

Python, OpenCV, Haar Cascade, PatchCore (Anomalib), PyTorch, Git, Raspberry Pi

- Designed and implemented a real-time video processing pipeline for object detection in live camera feeds. Integrated Haar Cascade classifiers and tuned preprocessing parameters for reliable detection.
- **Extended the system with a deep learning-based anomaly detection module using PatchCore (via Anomalib, built on PyTorch) to classify whether a bed was made or unmade.** Built and deployed an end-to-end inference pipeline, including dataset collection, model training, feature extraction, and threshold calibration for anomaly scoring.
- Optimized model inference for edge deployment, adapting to Raspberry Pi resource constraints. Implemented video stream encryption for secure transmission and storage.
- Structured the system into modular components (data ingestion, preprocessing, inference, alerting), improving maintainability, debugging, and testing.

Bird Dropper – Cloud-Ready Bird Tracking Web Application

JavaScript, Node.js, PostgreSQL, Docker, Render, Git

- Developed a modular full-stack web application for tracking bird sightings with user accounts, photo uploads, and interactive feeds.
 - Designed and implemented REST APIs for scalable backend services.
 - Containerized with Docker and deployed to Render, demonstrating cloud-native architecture.
 - Implemented secure authentication and cloud media storage for reliability and maintainability.
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Skills

Programming Languages: C++, Python, Java, JavaScript, HTML, CSS

Software Development & Tools: Git, Docker, REST APIs, Modular Design, Unit Testing, Cloud Deployment, Raspberry Pi

Other Tools & Technologies: Excel, OpenCV, Scikit-Learn, MATLAB, Matplotlib, AutoCAD